

**HERO scorecard (vertical) — value $\pm$ BCa 95% CI · colour = per-metric min-max (green = best) · * = Δ -vs-frozen CI excludes 0
9 metrics $\times$ 8 encoders · WINNER col + BLUE BOX = surgery-vs-pretrain champion duel (tie = paired 95% CI overlaps 0) · 10 K bootstrap**

	G-frozen	G-pre-2X	G-pre-enc	G-pre-hd	G-s3DI-enc	G-s3DI-hd	G-sNoDI-enc	G-sNoDI-hd	WINNER
action_top1 $\uparrow$ better	44.384 ± 2.274	53.151 $\pm 2.274^*$	51.671 $\pm 2.274^*$	42.849 ± 2.280	50.301 $\pm 2.274^*$	51.233 $\pm 2.301^*$	51.781 $\pm 2.274^*$	51.726 $\pm 2.301^*$	tie
motion_cos $\uparrow$ better	0.009 ± 0.001	0.169 $\pm 0.004^*$	0.159 $\pm 0.004^*$	0.013 $\pm 0.001^*$	0.163 $\pm 0.004^*$	0.167 $\pm 0.005^*$	0.168 $\pm 0.005^*$	0.179 $\pm 0.005^*$	ViT-G surgery noDI head
taxonomy_f1 $\uparrow$ better	0.793 ± 0.016	0.786 ± 0.017	0.791 ± 0.017	0.793 ± 0.017	0.793 ± 0.017	0.789 ± 0.017	0.790 ± 0.017	0.788 ± 0.017	tie
future_mse $\downarrow$ better	0.557 ± 0.001	0.546 $\pm 0.001^*$	0.540 $\pm 0.001^*$	0.558 $\pm 0.001^*$	0.515 $\pm 0.001^*$	0.540 $\pm 0.001^*$	0.515 $\pm 0.001^*$	0.540 $\pm 0.001^*$	ViT-G surgery 3stage_DI encoder
rollout $\downarrow$ better	0.009 ± 0.000	0.009 $\pm 0.000^*$	0.008 $\pm 0.000^*$	0.009 ± 0.000	0.009 $\pm 0.000^*$	0.008 $\pm 0.000^*$	0.009 $\pm 0.000^*$	0.008 $\pm 0.000^*$	tie
causal $\downarrow$ better	0.583 ± 0.001	0.571 $\pm 0.001^*$	0.565 $\pm 0.001^*$	0.583 ± 0.001	0.547 $\pm 0.002^*$	0.565 $\pm 0.001^*$	0.545 $\pm 0.002^*$	0.565 $\pm 0.001^*$	ViT-G surgery noDI encoder
tdist $\downarrow$ better	0.007 ± 0.000	0.008 $\pm 0.000^*$	0.007 ± 0.000	0.007 ± 0.000	0.008 $\pm 0.000^*$	0.007 ± 0.000	0.008 $\pm 0.000^*$	0.007 ± 0.000	tie
teacher_free $\downarrow$ better	0.046 ± 0.000	0.044 $\pm 0.001^*$	0.043 $\pm 0.001^*$	0.046 ± 0.000	0.048 $\pm 0.001^*$	0.043 $\pm 0.001^*$	0.047 $\pm 0.001^*$	0.043 $\pm 0.001^*$	tie
maskratio $\downarrow$ better	0.072 ± 0.001	0.071 $\pm 0.001^*$	0.068 $\pm 0.001^*$	0.072 ± 0.001	0.059 $\pm 0.001^*$	0.068 $\pm 0.001^*$	0.062 $\pm 0.001^*$	0.068 $\pm 0.001^*$	ViT-G surgery 3stage_DI encoder